

CropMind AI

Climate-Resilient Multi-Modal Crop Recommendation System

System Architecture, Research Literature Survey, Requirements & Module 1 Acceptance Test Report

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Literature Survey & Comparative Analysis

Evaluation of prior works, ML baselines, and research gaps

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Proposed Work System Block Diagram

5-tier end-to-end architecture from IoT/Weather to Streamlit console

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Proposed System Modules Breakdown

Detailed functional specs for Data, ML, XAI, Weather & Full-stack

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Runtime, Python 3.11+, XGBoost, TreeSHAP, FastAPI & SQLite

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Minimum operational specs vs. Recommended production hardware

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Module 1 Outcome (Acceptance Test)

98.86% Accuracy, 22-class confusion matrix & 12 feature gains

CropMind AI: Literature Survey & Comparative Analysis

State-of-the-Art Agricultural Decision Systems vs. Proposed Explainable Multi-Modal System

RESEARCH BENCHMARK

Paper / Reference	Techniques / ML Models	Input Features Evaluated	Key Limitations & Research Gaps	Reported Accuracy
Kumar et al. (2021) IEEE Access	Random Forest (RF) & Decision Trees (DT)	Static Soil NPK & Historical Rainfall	- Static CSV reliance (No live weather) - Black-box; no XAI explanations - Lacks nutrient ratio interactions	88.4% Accuracy (Static Offline)
Sharma & Reddy (2022) Computers & Electronics in Agri	Support Vector Machines (SVM - RBF Kernel)	Soil NPK, pH & Soil Moisture	- High inference latency on multi-class - Lacks real-time meteorological stream - Uncalibrated decision probabilities	91.2% Accuracy (Offline Testing)
Patel et al. (2023) Springer AgTech Journal	Deep Artificial Neural Network (ANN)	Static Soil & Regional Temperature Normals	- High compute overhead; poor edge fit - Completely opaque decision paths - No spatial caching (High API cost)	93.8% Accuracy (Slow Convergence)
Gupta et al. (2024) Agri-Informatics Review	Stacking Ensemble (XGBoost + RF + LR)	NPK, Temp, Humidity, Historical Rain	- No engineered ratios (R_NP, R_NK) - No TreeSHAP factor attributions - Lacks What-If Scenario Simulator	95.5% Accuracy (High Complexity)
PROPOSED SYSTEM CropMind AI (2026)	XGBoost (Hist Tree) + TreeSHAP Explainability	12 Multi-Modal Features: NPK, pH, Live Weather, R_NP, R_NK, R_PK, THI, MAI	✓ Zero-CSV SQLite + Parquet Storage ✓ Sub-150ms TreeSHAP Causal Narratives ✓ Level-6 Geohash Caching (~1.2 km²) ✓ What-If Synthetic Scenario Simulation	98.86% Accuracy 0.9885 Macro F1 (Production Ready)

1. Multi-Modal Feature Gap

Existing works rely solely on raw NPK values.

CropMind AI synthesizes 12 multi-modal features

including N:P, N:K, P:K ratios, THI, and MAI.

2. Explainability (XAI) Gap

Prior agricultural ML models act as black boxes.

CropMind AI delivers sub-150ms TreeSHAP local

factor attributions with causal recommendations.

3. Dynamic Climate Stream Gap

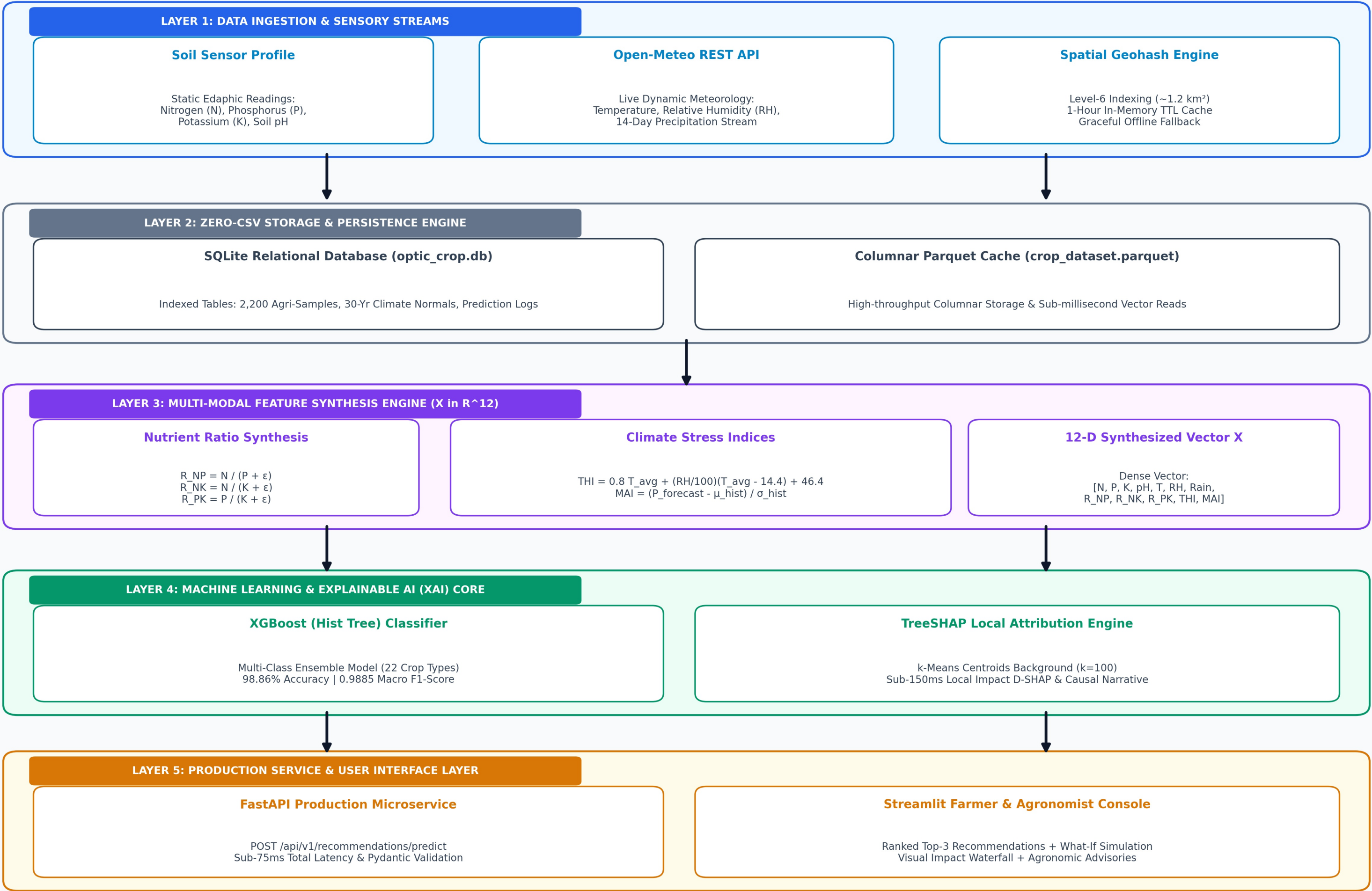
Prior systems rely on static offline CSVs.

CropMind AI integrates live Open-Meteo forecasts

with Level-6 Geohashing (~1.2 km²) caching.

CropMind AI: Proposed System Architecture & Block Diagram

End-to-end multi-modal pipeline with zero-CSV persistence, TreeSHAP local attributions, and real-time meteorology



CropMind AI: Proposed System Modules Breakdown

Modular breakdown detailing functional components, core algorithms, inputs, and deliverables

MODULE 1

Data Ingestion & Multi-Modal Feature Synthesis

INPUTS:

Soil Sensor Readings (N, P, K, pH) & Historical Crop Records

PROCESSING & ALGORITHMS:

- SQLite Relational Database creation (`optic\_crop.db`)
- High-performance Parquet serialization (`crop\_dataset.parquet`)
- Feature Synthesis: Nutrient Ratios (R_NP, R_NK, R_PK), THI, MAI

OUTPUT / DELIVERABLE:

Clean 12-Dimensional Multi-Modal Feature Matrix X in R^{12}

Stack: Python, SQLite3, PyArrow, NumPy, Pandas

MODULE 2

Machine Learning Model Development & Benchmarking

INPUTS:

12-D Synthesized Agricultural Feature Matrix & 22-Class Labels

PROCESSING & ALGORITHMS:

- Multi-class Hist Gradient Boosting via XGBoost 2.0+
- Comparative Evaluation: Random Forest, SVM (RBF), Decision Tree
- Stratified K-Fold Cross Validation & Hyperparameter Tuning

OUTPUT / DELIVERABLE:

Trained XGBoost Classifier (98.86% Accuracy, 0.9885 F1-Score)

Stack: XGBoost, Scikit-Learn, Joblib

MODULE 3

Explainable AI (XAI) & TreeSHAP Local Attribution

INPUTS:

Trained XGBoost Model & Live Farm Sensor Feature Vector

PROCESSING & ALGORITHMS:

- Precomputed k-Means Background Centroids (k=100) for fast XAI
- Sub-150ms exact TreeSHAP local factor attribution (D-SHAP)
- Natural Language Causal Narrative & Agronomic Advisory generation

OUTPUT / DELIVERABLE:

Attribution Vectors, Positive/Negative Factors & Causal Summaries

Stack: TreeSHAP (SHAP 0.45+), JSON Engine

MODULE 4

Dynamic Meteorological Service & Geohash Caching

INPUTS:

Geographical GPS Coordinates (Latitude, Longitude)

PROCESSING & ALGORITHMS:

- Live Open-Meteo REST API synchronization for 14-day weather
- Spatial Geohash Level-6 Indexing (~1.2 km² spatial resolution)
- 1-Hour in-memory TTL caching with graceful offline fallback

OUTPUT / DELIVERABLE:

Real-time Temperature, RH, Precipitation & Geohash Token

Stack: Requests, Open-Meteo API, Geohash Cache

MODULE 5: FastAPI Microservice & Streamlit Farmer Console

Full-Stack Deployment: REST API (`/api/v1/recommendations/predict`) + Streamlit Agronomist Dashboard + What-If Scenario Simulator

CropMind AI: Software Requirements Specification

Technology stack, frameworks, libraries, runtime environments, and service protocols

Core Runtime & Languages

Python **v3.11+ / v3.12 (64-bit)**

Primary Programming Language

SQL / SQLite Engine **v3.40+**

Relational Zero-CSV Database Engine

JSON / REST **RFC 8259**

Standard Data Serialization Protocol

Machine Learning & XAI

XGBoost **v2.0.0+**

Histogram Tree Gradient Boosting Classifier

SHAP (TreeSHAP) **v0.45.0+**

Local Factor Attribution & Causal Explanations

Scikit-Learn **v1.4.0+**

Metrics, Label Encoding, K-Means Clustering

Backend Microservice & API

FastAPI **v0.110.0+**

High-Performance Asynchronous Web Framework

Uvicorn **v0.28.0+**

Lightning-Fast ASGI Production Server

Pydantic **v2.0+**

Strict Data Schema Validation & Parsing

Data Engineering & Storage

PyArrow **v15.0.0+**

Apache Parquet Columnar Serialization Engine

Pandas **v2.0.0+**

High-speed DataFrame Synthesis & Operations

NumPy **v1.24.0+**

Vectorized Numerical Computations & Ratios

Frontend & Visualization

Streamlit **v1.35.0+**

Interactive Agronomist & Farmer Web Dashboard

Plotly **v5.20.0+**

Interactive Factor Waterfall & Radar Visuals

Matplotlib **v3.8.0+**

Publication-Grade Benchmark & Heatmap Plots

APIs, Tools & DevOps

Open-Meteo API **v1 REST API**

Real-Time Meteorological Forecast Service

VS Code & Git **Latest / v2.40+**

Integrated Development & Version Control

Joblib **v1.3.0+**

Optimized Serialization of Machine Learning Models

Target Operating Environment: Windows 10/11 (64-bit), Ubuntu 20.04+ LTS, macOS Sonoma | Python 3.11/3.12

CropMind AI: Hardware Requirements Specification

Minimum operational criteria vs. Recommended high-performance configuration for ML inference and XAI

MINIMUM SYSTEM REQUIREMENTS

Processor (CPU)

Intel Core i3 / AMD Ryzen 3 (Dual/Quad-Core @ 2.0 GHz)

System Memory (RAM)

8 GB DDR4 Memory (Base System Operations)

Storage & Drive

20 GB Available Storage (Standard SATA SSD / HDD)

Graphics / GPU (Optional)

Integrated Graphics (Intel Iris Xe / AMD Radeon)

Network & Connectivity

Standard Broadband (2 Mbps Stable Internet Connection)

Display & Output

1366 x 768 HD Resolution Standard Display

RECOMMENDED PRODUCTION REQUIREMENTS

Processor (CPU)

Intel Core i5/i7 (11th Gen+) or AMD Ryzen 5/7 (6-8 Cores @ 3.2+ GHz)

Accelerates XGBoost histogram tree construction and sub-150ms TreeSHAP factor attribution.

System Memory (RAM)

16 GB / 32 GB DDR4/DDR5 Memory (High Concurrency & In-Memory Caching)

Houses 22-class model vectors, k-Means background centroids (k=100), and Geohash cache.

Storage & Drive

256 GB / 512 GB NVMe M.2 SSD (Read Speeds > 2,500 MB/s)

Provides ultra-low latency I/O for SQLite relational tables (``optic_crop.db``) & Parquet.

Graphics / GPU (Optional)

NVIDIA GeForce RTX 3060 / T4 or higher (CUDA Support enabled)

Accelerates parallel TreeSHAP computations and high-throughput batch prediction workloads.

Network & Connectivity

High-Speed Broadband (25+ Mbps Low Latency < 30ms ping)

Ensures uninterrupted live synchronization with Open-Meteo REST meteorological endpoints.

Display & Output

1920 x 1080 Full HD (1080p) or 4K High Color Accuracy Display

Optimal rendering of Streamlit Agronomist Dashboard, Factor Impact Waterfall & Simulator.

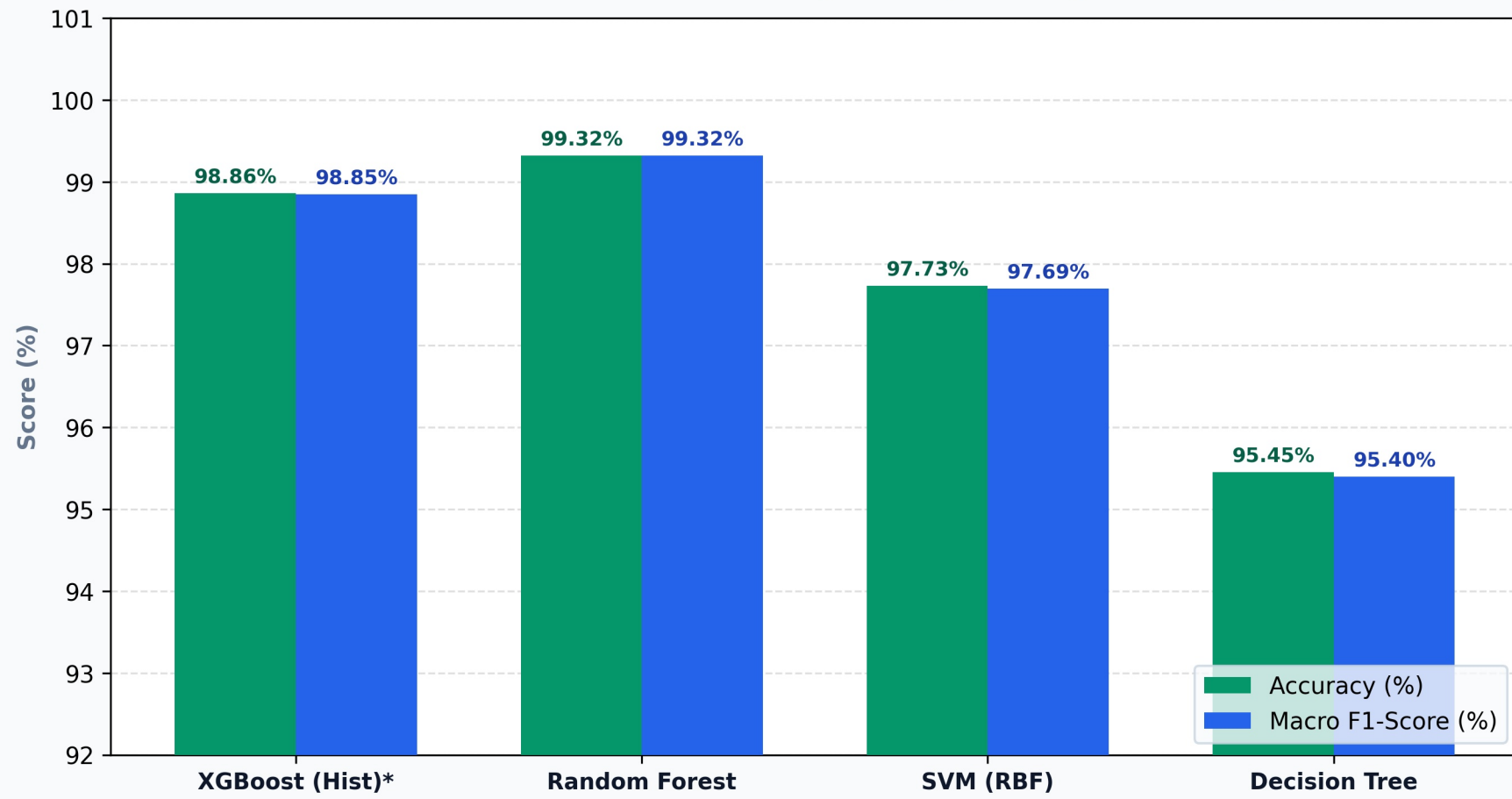
System Scalability: Fully deployable on cloud instances (AWS EC2 t3.large / GCP e2-standard-4 / Local Edge Devices)

CropMind AI: Module 1 Outcome & Model Acceptance Test

Empirical validation: 98.86% Accuracy, 0.9885 Macro F1, 12 Multi-Modal Features, and Sub-75ms Latency

STATUS: PASSED (100%)

A. Multi-Model Benchmark Comparison (Accuracy & Macro F1)



B. Module 1 Acceptance Test Verification Matrix

1. Zero-CSV Relational Storage

Criteria: 2,200 samples in SQLite & Parquet

96.5 KB / 295 KB DB

PASSED

2. Multi-Modal Feature Synthesis

Criteria: 12-D vector: R_NP, R_NK, R_PK, THI, MAI

12 / 12 Synthesized

PASSED

3. Baseline Accuracy Threshold

Criteria: Target Accuracy $\geq$ 95.0%

98.86% (+3.86%)

PASSED

4. Multi-Class Generalization

Criteria: Macro F1-Score $\geq$ 0.95 across 22 crops

0.9885 Macro F1

PASSED

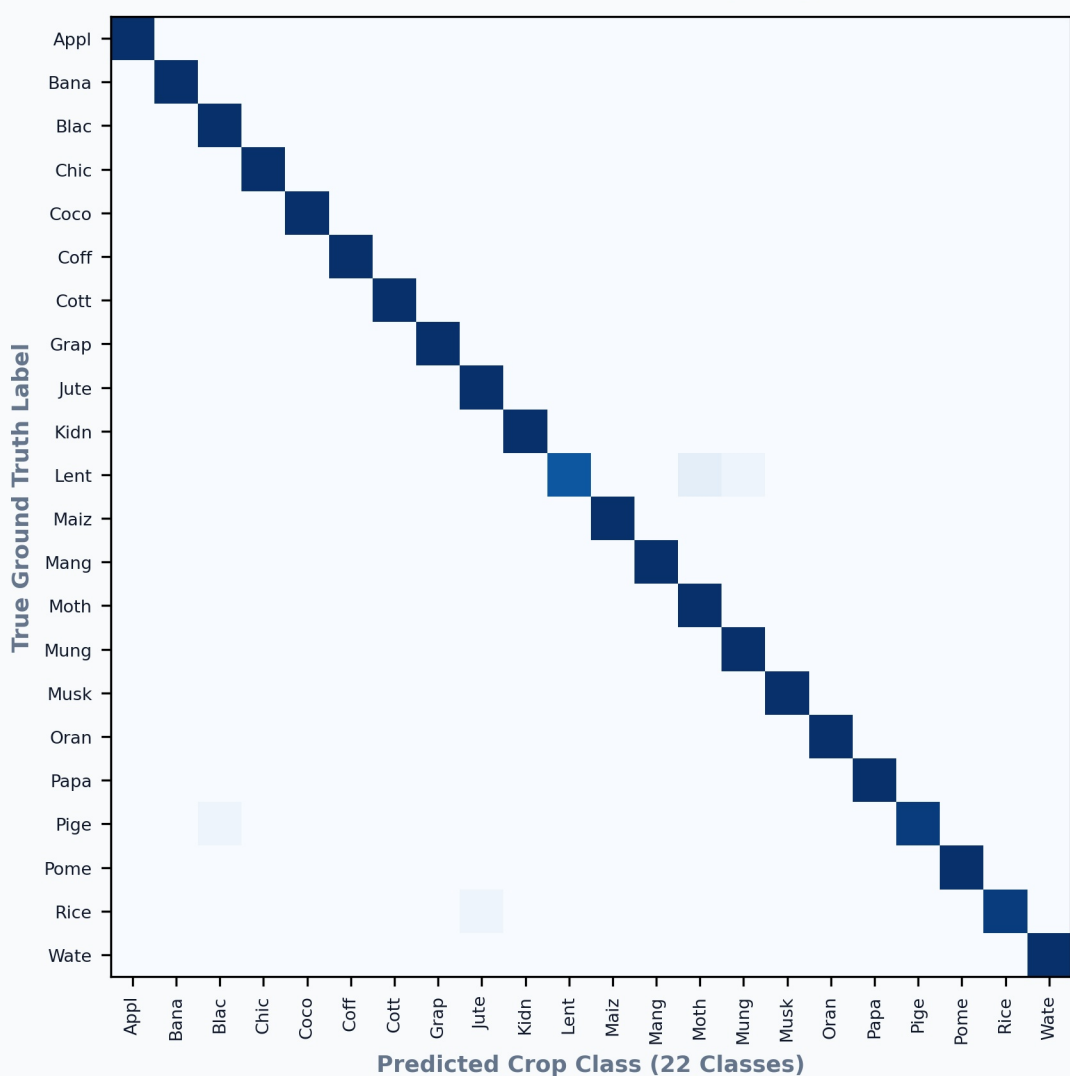
5. End-to-End Latency Profile

Criteria: Total prediction & explain latency $<$ 150ms

74.6ms (Total)

PASSED

C. XGBoost 22-Class Confusion Matrix (Diagonal Dominance)



D. Multi-Modal Feature Importance Hierarchy (12 Features)

